PAPER ID-410086

Subject Code: RCS502

Roll No: | | | | | | | | | | | | | |

B. TECH.
(SEM V) THEORY EXAMINATION 2021-22
DESIGN & ANALYSIS OF ALGORITHM

Time: 3 Hours ***Total Marks: 70***

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 7 = 14

a. Explain Big O notation in brief.
b. Write an Insertion sort algorithm.
c. Write the properties of B-Trees.
d. How insert a node in skip list?
e. How Backtracking approach is different from branch & bound approach?
f. Explain the elements of Dynamic programming.
g. What is NP-Hard problem?

SECTION B

2. Attempt any *three* of the following: 7 x 3 = 21

a. Write Quick sort algorithm and sort the following sequence (12, 55, 44, 21, 87, 16) using quick sort.
b. Explain Insertion operation in Red Black tree with suitable example.
c. Define minimum spanning tree (MST). Write Prim's algorithm to generate a MST for any given weighted graph.
d. What is N queens problem? How N queens problem is solved by backtracking.
e. Explain and Write the Knuth-Morris-Pratt algorithm for pattern matching with suitable example.

SECTION C

3. Attempt any *one* part of the following: 7 x 1 = 7

(a) Illustrate the operation of counting sort on the following array: **A={6, 0, 2, 0, 1, 3, 4, 6, 1, 3, 2}**
(b) Solve the recurrence using recursion tree method
$T(n) = 3T(n/4) + cn^2$

4. Attempt any *one* part of the following: 7 x 1 = 7

(a) What is skip list? Explain the Insertion operation in Skip list with suitable example.
(b) Insert the following list of data items 20, 30, 35, 85, 10, 55, 60, 25, 5, 65, 70, 75, 15, 40, 50, 80, 45 in an empty B-tree. Using minimum degree't' as 3.

5. Attempt any *one* part of the following: 7 x 1 = 7

(a) Write down the Dijkstra algorithm to solve the single source shortest path problem also write its time complexity.
(b) What do you understand by convex hull? Discuss the algorithm to solve the convex hull problem.

6. Attempt any *one* part of the following: 7 x 1 = 7

(a) Explain the method of finding Hamiltonian cycles in a graph using backtracking method with suitable example.
(b) Describe the Floyd Warshall algorithm for finding the all pair shortest path problem with suitable example

7. Attempt any *one* part of the following: 7 x 1 = 7

(a) Write and Explain Boyer-Moore Algorithm for string matching algorithm.
(b) Prove that APPROX- VERTEX-COVER is a polynomial time 2-approximation algorithm